Supplementary Material: Recovering Stranded Discrimination in Knowledge Tracing

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1 Per-Backbone Results

Tables 1 and 2 report AUC and NLL for each backbone individually (mean ± std across 3 seeds). Per-backbone results show that the AUC gain is broad across architectures, with AKT on Eedi (−0.19 pp) as the only exception; there, AKT's item-attention mechanism already captures most per-item variation, leaving little headroom for post-hoc correction. SAKT consistently shows the largest gains (+2.98–6.17 pp), consistent with its weaker per-item modeling capacity.

2 Classic Score-Only Calibrators

Table 3 makes the score-only comparison visible: temperature scaling matches Platt in AUC to four decimals on all four datasets, isotonic regression changes AUC only negligibly, and histogram binning can lower AUC through ties. These results are averaged over the same five backbones as the main-text tables.

3 Sensitivity to Prior Variance σ_b^2

The static SLC estimator (Eq. 6 in the main text) uses a fixed prior variance $\sigma_b^2 = 1.0$ for the per-item bias. This appendix reports a controlled sensitivity sweep over $\sigma_b^2 \in \{0.01, 0.1, 0.5, 1.0, 5.0, 10.0, 100.0\}$ and explains why the result is backbone-independent.

We load the AS17 calibration and test splits (188 554 and 188 565 tokens, 3 162 items; median 48 observations per item). A constant backbone $p_0 = \bar{y}_{\text{calib}} \approx 0.55$ replaces the trained KT model. For each σ_b^2 value we: (i) compute the static per-item bias $\hat{b}_i$ on the calibration set via Eq. 6, (ii) fit the two-parameter Offset-Platt link (a^*, b_0^*) by IRLS on the same calibration set, and (iii) evaluate AUC and NLL on the held-out test set.

A constant backbone is sufficient for this sweep because the sensitivity of $\hat{b}_i$ to σ_b^2 is governed entirely by the shrinkage fraction $\lambda_i = W_i/(1/\sigma_b^2 + W_i)$, where $W_i = \sum_n p_n(1-p_n)$ is the total Fisher-information weight of item i. For

Table 1. Per-backbone AUC (↑), mean ± std across 3 seeds. Best per column in **bold**.

Dataset	Method	AKT	DKT	SAKT	DKVMN	LPKT
AS17	Base	0.7189 ± 0.003	0.6991 ± 0.000	0.6193 ± 0.001	0.6841 ± 0.001	0.6855 ± 0.000
	Platt	0.7189 ± 0.003	0.6991 ± 0.000	0.6193 ± 0.001	0.6841 ± 0.001	0.6855 ± 0.000
	Platt-T	0.7198 ± 0.003	0.6991 ± 0.000	0.6193 ± 0.001	0.6841 ± 0.001	0.6855 ± 0.000
	ResCal	0.7275 ± 0.002	0.7222 ± 0.001	0.6613 ± 0.001	0.7096 ± 0.000	0.7117 ± 0.000
	ResCal+Iso	0.7274 ± 0.002	0.7221 ± 0.001	0.6613 ± 0.001	0.7094 ± 0.000	0.7118 ± 0.000
	Naive	0.7060 ± 0.002	0.7079 ± 0.001	0.6584 ± 0.001	0.6984 ± 0.000	0.7022 ± 0.000
	SLC	$\mathbf{0.7321} \pm 0.002$	$\mathbf{0.7325} \pm 0.001$	$\mathbf{0.6811} \pm 0.001$	$\mathbf{0.7214} \pm 0.000$	$\mathbf{0.7237} \pm 0.000$
	SLC-T	0.7303 ± 0.002	0.7312 ± 0.001	0.6791 ± 0.000	0.7201 ± 0.000	0.7222 ± 0.000
	SLC+Iso	0.7314 ± 0.002	0.7314 ± 0.001	0.6790 ± 0.000	0.7204 ± 0.000	0.7225 ± 0.000
Eedi	Base	0.7733 ± 0.000	0.7207 ± 0.000	0.7196 ± 0.000	0.7219 ± 0.000	0.6591 ± 0.000
	Platt	0.7733 ± 0.000	0.7207 ± 0.000	0.7196 ± 0.000	0.7219 ± 0.000	0.6591 ± 0.000
	Platt-T	0.7733 ± 0.000	0.7207 ± 0.000	0.7196 ± 0.000	0.7219 ± 0.000	0.6594 ± 0.000
	ResCal	0.7734 ± 0.000	0.7589 ± 0.000	0.7568 ± 0.000	0.7592 ± 0.000	0.7188 ± 0.000
	ResCal+Iso	0.7734 ± 0.000	0.7589 ± 0.000	0.7568 ± 0.000	0.7592 ± 0.000	0.7188 ± 0.000
	Naive	0.7509 ± 0.000	0.7433 ± 0.000	0.7433 ± 0.000	0.7447 ± 0.000	0.7051 ± 0.000
	SLC	0.7714 ± 0.000	$\mathbf{0.7643} \pm 0.000$	$\mathbf{0.7628} \pm 0.000$	$\mathbf{0.7650} \pm 0.000$	$\mathbf{0.7259} \pm 0.000$
	SLC-T	0.7704 ± 0.000	0.7637 ± 0.000	0.7622 ± 0.000	0.7644 ± 0.000	0.7255 ± 0.000
	SLC+Iso	0.7709 ± 0.000	0.7634 ± 0.000	0.7622 ± 0.000	0.7643 ± 0.000	0.7254 ± 0.000
AS09	Base	0.6925 ± 0.009	0.6987 ± 0.005	0.6084 ± 0.012	0.6817 ± 0.007	0.7099 ± 0.001
	Platt	0.6925 ± 0.009	0.6987 ± 0.005	0.6084 ± 0.012	0.6817 ± 0.007	0.7099 ± 0.001
	Platt-T	0.6918 ± 0.008	0.6994 ± 0.004	0.6113 ± 0.012	0.6820 ± 0.007	0.7087 ± 0.002
	ResCal	0.6928 ± 0.009	0.6989 ± 0.005	0.6085 ± 0.012	0.6819 ± 0.007	0.7102 ± 0.001
	ResCal+Iso	0.6922 ± 0.008	0.6986 ± 0.005	0.6085 ± 0.013	0.6819 ± 0.007	0.7101 ± 0.001
	Naive	0.6712 ± 0.006	0.6706 ± 0.002	0.6316 ± 0.003	0.6686 ± 0.002	0.6847 ± 0.001
	SLC	$\mathbf{0.7131} \pm 0.011$	$\mathbf{0.7198} \pm 0.004$	$\mathbf{0.6570} \pm 0.003$	$\mathbf{0.7078} \pm 0.005$	$\mathbf{0.7355} \pm 0.002$
	SLC-T	0.7089 ± 0.010	0.7147 ± 0.004	0.6532 ± 0.003	0.7035 ± 0.005	0.7300 ± 0.002
	SLC+Iso	0.7108 ± 0.010	0.7123 ± 0.003	0.6512 ± 0.004	0.7033 ± 0.005	0.7295 ± 0.001
Algebra	Base	0.8168 ± 0.000	0.8254 ± 0.000	0.7918 ± 0.001	0.8179 ± 0.000	0.8158 ± 0.001
	Platt	0.8168 ± 0.000	0.8254 ± 0.000	0.7918 ± 0.001	0.8179 ± 0.000	0.8158 ± 0.001
	Platt-T	0.8169 ± 0.000	0.8254 ± 0.000	0.7918 ± 0.001	0.8179 ± 0.000	0.8158 ± 0.001
	ResCal	0.8204 ± 0.001	0.8344 ± 0.000	0.8094 ± 0.001	0.8290 ± 0.000	0.8281 ± 0.001
	ResCal+Iso	0.8204 ± 0.001	0.8343 ± 0.000	0.8095 ± 0.001	0.8290 ± 0.000	0.8281 ± 0.001
	Naive	0.7947 ± 0.001	0.8206 ± 0.000	0.7998 ± 0.002	0.8167 ± 0.001	0.8168 ± 0.000
	SLC	$\mathbf{0.8245} \pm 0.001$	$\mathbf{0.8417} \pm 0.000$	$\mathbf{0.8216} \pm 0.001$	$\mathbf{0.8377} \pm 0.000$	$\mathbf{0.8372} \pm 0.000$
	SLC-T	0.8225 ± 0.001	0.8408 ± 0.000	0.8206 ± 0.001	0.8367 ± 0.000	0.8362 ± 0.000
	SLC+Iso	0.8242 ± 0.001	0.8409 ± 0.000	0.8209 ± 0.001	0.8369 ± 0.000	0.8365 ± 0.000

a constant backbone, $W_i = n_i \cdot p_0(1-p_0) \approx 0.2475\, n_i$; for any trained backbone with predictions spread over $[0, 1]$, the per-token weight $p_n(1-p_n) \in [0, 0.25]$ averages to $\approx$ 0.20–0.24, so W_i differs by $< 15\%$. Because λ_i is monotone in W_i and saturates rapidly for $W_i \gg 1/\sigma_b^2$, the sensitivity pattern is backbone-invariant. The constant backbone yields a slightly *higher* W_i (since $p(1-p)$ is maximized at $p = 0.5$), making this a conservative test: a trained backbone would show even less sensitivity.

Table 4 reports the sweep. Over the two-order-of-magnitude range $\sigma_b^2 \in [0.1, 10]$, AUC varies by < 0.4 pp and NLL by < 0.004. Only the extreme $\sigma_b^2 = 0.01$ (prior precision = 100, equivalent to heavy shrinkage toward zero) shows a visible AUC loss of $\sim$0.9 pp, consistent with excessive regularization suppressing genuine item biases.

Table 2. Per-backbone NLL (↓), mean ± std across 3 seeds. Best per column in **bold**.

Dataset	Method	AKT	DKT	SAKT	DKVMN	LPKT
AS17	Base	0.603 ± 0.003	0.607 ± 0.001	0.651 ± 0.001	0.617 ± 0.001	0.615 ± 0.000
	Platt	0.593 ± 0.002	0.605 ± 0.000	0.644 ± 0.000	0.614 ± 0.000	0.612 ± 0.000
	Platt-T	0.592 ± 0.002	0.604 ± 0.000	0.643 ± 0.000	0.613 ± 0.000	0.611 ± 0.000
	ResCal	0.595 ± 0.003	0.591 ± 0.001	0.630 ± 0.001	0.601 ± 0.001	0.599 ± 0.000
	ResCal+Iso	0.587 ± 0.002	0.590 ± 0.001	0.627 ± 0.001	0.599 ± 0.000	0.598 ± 0.000
	Naive	0.662 ± 0.005	0.635 ± 0.004	0.666 ± 0.001	0.642 ± 0.002	0.635 ± 0.001
	SLC	$\mathbf{0.585} \pm 0.002$	$\mathbf{0.583} \pm 0.001$	$\mathbf{0.616} \pm 0.000$	$\mathbf{0.592} \pm 0.000$	$\mathbf{0.590} \pm 0.000$
	SLC-T	0.587 ± 0.002	0.585 ± 0.001	0.618 ± 0.000	0.594 ± 0.000	0.591 ± 0.000
	SLC+Iso	0.585 ± 0.002	0.585 ± 0.001	0.618 ± 0.000	0.594 ± 0.000	0.592 ± 0.000
Eedi	Base	0.540 ± 0.000	0.580 ± 0.000	0.582 ± 0.001	0.579 ± 0.000	0.612 ± 0.000
	Platt	0.539 ± 0.000	0.580 ± 0.000	0.581 ± 0.000	0.579 ± 0.000	0.612 ± 0.000
	Platt-T	0.539 ± 0.000	0.580 ± 0.000	0.581 ± 0.000	0.579 ± 0.000	0.611 ± 0.000
	ResCal	0.540 ± 0.000	0.552 ± 0.000	0.554 ± 0.001	0.552 ± 0.000	0.581 ± 0.000
	ResCal+Iso	0.539 ± 0.000	0.552 ± 0.000	0.553 ± 0.000	0.551 ± 0.000	0.580 ± 0.000
	Naive	0.574 ± 0.001	0.573 ± 0.000	0.576 ± 0.001	0.574 ± 0.000	0.599 ± 0.000
	SLC	0.541 ± 0.000	$\mathbf{0.548} \pm 0.000$	$\mathbf{0.549} \pm 0.000$	$\mathbf{0.548} \pm 0.000$	$\mathbf{0.576} \pm 0.000$
	SLC-T	0.542 ± 0.000	0.549 ± 0.000	0.550 ± 0.000	0.548 ± 0.000	0.576 ± 0.000
	SLC+Iso	0.542 ± 0.000	0.548 ± 0.000	0.549 ± 0.000	0.547 ± 0.000	0.576 ± 0.000
AS09	Base	0.630 ± 0.035	0.590 ± 0.004	0.710 ± 0.012	0.629 ± 0.020	0.579 ± 0.003
	Platt	0.593 ± 0.017	0.577 ± 0.004	0.625 ± 0.004	0.598 ± 0.009	0.569 ± 0.002
	Platt-T	0.593 ± 0.016	0.575 ± 0.004	0.621 ± 0.002	0.596 ± 0.008	0.567 ± 0.002
	ResCal	0.629 ± 0.035	0.589 ± 0.004	0.708 ± 0.012	0.629 ± 0.020	0.578 ± 0.003
	ResCal+Iso	0.594 ± 0.017	0.577 ± 0.004	0.625 ± 0.004	0.598 ± 0.007	0.571 ± 0.002
	Naive	0.781 ± 0.069	0.702 ± 0.010	0.917 ± 0.055	0.758 ± 0.028	0.694 ± 0.004
	SLC	0.599 ± 0.025	$\mathbf{0.575} \pm 0.006$	0.636 ± 0.009	0.606 ± 0.015	$\mathbf{0.563} \pm 0.003$
	SLC-T	0.608 ± 0.027	0.582 ± 0.006	0.649 ± 0.011	0.617 ± 0.017	0.570 ± 0.003
	SLC+Iso	0.622 ± 0.029	0.610 ± 0.008	0.653 ± 0.007	0.640 ± 0.011	0.590 ± 0.003
Algebra	Base	0.348 ± 0.006	0.328 ± 0.001	0.362 ± 0.003	0.335 ± 0.000	0.335 ± 0.000
	Platt	0.340 ± 0.003	0.327 ± 0.001	0.357 ± 0.001	0.334 ± 0.001	0.334 ± 0.000
	Platt-T	0.340 ± 0.002	0.327 ± 0.001	0.357 ± 0.001	0.334 ± 0.001	0.334 ± 0.000
	ResCal	0.345 ± 0.006	0.322 ± 0.001	0.350 ± 0.003	0.327 ± 0.000	0.326 ± 0.000
	ResCal+Iso	0.331 ± 0.001	0.320 ± 0.001	0.339 ± 0.001	0.325 ± 0.001	0.325 ± 0.000
	Naive	0.557 ± 0.011	0.410 ± 0.002	0.462 ± 0.025	0.418 ± 0.002	0.410 ± 0.003
	SLC	$\mathbf{0.336} \pm 0.003$	$\mathbf{0.316} \pm 0.001$	$\mathbf{0.339} \pm 0.002$	$\mathbf{0.320} \pm 0.001$	$\mathbf{0.319} \pm 0.000$
	SLC-T	0.340 ± 0.003	0.317 ± 0.001	0.342 ± 0.002	0.322 ± 0.001	0.321 ± 0.000
	SLC+Iso	0.335 ± 0.002	0.321 ± 0.001	0.340 ± 0.001	0.325 ± 0.001	0.324 ± 0.000

4 Connection Between the Static Correction Block and Ridge Logistic Regression

We show that the iterative version of the static per-item correction block is exactly the corresponding IRLS procedure for ℓ_2-penalized logistic regression with per-item intercepts and the backbone logit as a fixed offset. This clarifies the objective behind Eq. 6 in the main text. The deployed SLC in Algorithm 1 uses the associated single-pass blockwise estimate, followed by a separate offset-Platt fit.

4.1 Setup

Consider N observations with binary labels $y_n \in \{0, 1\}$, frozen backbone logits $\eta_{0,n}$, and item assignments $i_n \in \{1, \ldots, K\}$. The per-item correction model is $p_n = \sigma(\eta_{0,n} + b_{i_n})$, where σ denotes the sigmoid function. The ℓ_2-penalized

Table 3. Classic score-only calibrators, averaged over 5 backbones at full calibration fraction. Temperature scaling matches Platt in AUC; isotonic regression changes AUC only marginally; histogram binning can reduce AUC through ties.

	Algebra		AS17		Eedi		AS09	
Method	AUC	NLL	AUC	NLL	AUC	NLL	AUC	NLL
Base	0.8135	0.341	0.6814	0.618	0.7189	0.579	0.6782	0.628
Platt	0.8135	0.338	0.6814	0.613	0.7189	0.578	0.6782	0.592
TS	0.8135	0.339	0.6814	0.614	0.7189	0.578	0.6782	0.613
Iso	0.8135	0.335	0.6813	0.614	0.7189	0.578	0.6780	0.593
Hist	0.8006	0.340	0.6793	0.614	0.7162	0.579	0.6766	0.592

Table 4. Sensitivity of SLC to prior variance σ_b^2 on AS17 (constant backbone). The default $\sigma_b^2 = 1.0$ is shaded.

σ_b^2	AUC (↑)	ΔAUC (pp)	NLL (↓)	ΔNLL
0.01	0.6445	+14.45	0.6522	−0.0147
0.10	0.6504	+15.04	0.6348	−0.0321
0.50	0.6532	+15.32	0.6315	−0.0354
1.00	0.6537	+15.37	0.6315	−0.0354
5.00	0.6535	+15.35	0.6329	−0.0340
10.00	0.6534	+15.34	0.6338	−0.0331
100.00	0.6533	+15.33	0.6354	−0.0315
Span over $[0.1, 10]$: AUC = 0.33 pp, NLL = 0.003.				

negative log-likelihood (“ridge logistic regression”) is:

$$\mathcal{L}(\mathbf{b}) = -\sum_{n=1}^{N}\left[y_n \log p_n + (1-y_n)\log(1-p_n)\right] + \frac{\lambda}{2}\sum_{i=1}^{K} b_i^2, \qquad \lambda = 1/\sigma_b^2. \quad (1)$$

4.2 Hessian is Diagonal

The gradient of $\mathcal{L}$ with respect to b_i is:

$$\frac{\partial \mathcal{L}}{\partial b_i} = -\sum_{n:\, i_n = i} (y_n - p_n) + \lambda\, b_i \; = \; -g_i + \lambda\, b_i,$$

where $g_i \triangleq \sum_{n:\, i_n=i}(y_n - p_n)$ is the per-item score residual.

The Hessian entries are:

$$\frac{\partial^2 \mathcal{L}}{\partial b_i\, \partial b_j} = \begin{cases} W_i + \lambda & \text{if } i = j, \\ 0 & \text{if } i \neq j, \end{cases}$$

where $W_i = \sum_{n:\, i_n=i} p_n(1-p_n)$ is the Fisher information weight. **The off-diagonal is zero** because each observation n involves exactly one item i_n: the derivative $\partial p_n/\partial b_j = 0$ whenever $j \neq i_n$. Hence $H = \mathrm{diag}(W_1+\lambda, \ldots, W_K+\lambda)$.

4.3 Newton Step Decomposes into K Independent Scalar Updates

The Newton–Raphson update $\mathbf{b}^{(\ell+1)} = \mathbf{b}^{(\ell)} - H^{-1}\nabla\mathcal{L}$ reduces to K independent updates:

$$b_i^{(\ell+1)} = b_i^{(\ell)} + \frac{g_i - \lambda\, b_i^{(\ell)}}{W_i + \lambda} = \frac{W_i \cdot b_i^{(\ell)} + g_i}{W_i + 1/\sigma_b^2}. \tag{2}$$

4.4 Comparison with the Iterated Static Correction Block

The iterated static correction block computes:
1. $p_n = \sigma(\eta_{0,n} + b_{i_n}^{(\ell)})$ for all n,
2. $W_i = \sum_{n:\, i_n = i} p_n(1-p_n), \quad g_i = \sum_{n:\, i_n = i}(y_n - p_n)$,
3. $b_i^{(\ell+1)} = \frac{W_i \cdot b_i^{(\ell)} + g_i}{1/\sigma_b^2 + W_i}$.

This is identical to Eq. (2)**.** Given the same initialization $b_i^{(0)} = 0$, the iterated blockwise solver and ridge-logistic IRLS produce the same updates at every step and converge to the same fixed point.

> **Proposition (Blockwise ridge connection).** Under the conditions that (i) the backbone logit $\eta_{0,n}$ is a fixed offset, (ii) ℓ_2 penalty $\lambda = 1/\sigma_b^2$ is applied only to the item intercepts b_i, and (iii) initialization is $b_i^{(0)} = 0$, the repeated blockwise updates of the static correction block are identical to the IRLS iterates of the corresponding penalized logistic model. The converged block estimator is the posterior mean of a Gaussian random-intercept model under the Laplace approximation.

The iterated blockwise solver and ridge logistic converge to the same static correction solution, but differ in implementation. Standard ridge logistic regression constructs a design matrix of dimension $N \times K$ and solves a K-dimensional system at each Newton step, yielding $O(NK)$ cost per iteration. SLC exploits the diagonal Hessian structure by computing per-item `bincount` aggregates ($O(N)$) followed by K scalar divisions ($O(K)$), for a total cost of $O(N+K)$ per iteration. This decomposition also enables seamless extension to temporal smoothing via the Rauch–Tung–Striebel smoother at cost $O(K \cdot T)$.

4.5 Empirical Verification

To confirm the theoretical equivalence, we run a fully-converged ridge logistic regression baseline (50 IRLS iterations, same $\lambda{=}1/\sigma_b^2$, followed by the same offset-Platt link) alongside the single-pass SLC on two datasets and two backbones (3 seeds each). Table 5 reports the difference: the maximum discrepancy is 0.07 pp in AUC and 0.001 in NLL, both well within seed-level variance. The sign of ΔAUC is inconsistent across configurations, confirming that the residual gap is noise rather than a systematic bias from the single-pass approximation.

Table 5. Ridge logistic regression vs. SLC (single-pass). Δ = Ridge − SLC, averaged over 3 seeds (± std). All differences are within seed-level noise.

Dataset	Backbone	Ridge AUC	SLC AUC	ΔAUC	ΔNLL
AS17	AKT	0.7327 ± 0.0021	0.7321 ± 0.0025	+0.07 pp	−0.001
AS17	DKT	0.7329 ± 0.0010	0.7325 ± 0.0011	+0.04 pp	<0.001
AS09	AKT	0.7132 ± 0.0130	0.7131 ± 0.0130	+0.01 pp	−0.001
AS09	DKT	0.7195 ± 0.0044	0.7198 ± 0.0045	−0.03 pp	+0.001

If the backbone scale a is jointly optimized with b_i (as in standard sklearn `LogisticRegression` with per-item one-hot features), the Hessian acquires off-diagonal blocks $\partial^2\mathcal{L}/\partial a\,\partial b_i \neq 0$, and the global Newton step no longer decomposes. SLC therefore separates bias estimation (b_i only) from link estimation ((a, b_0) via offset-Platt), preserving the additive-offset parameterization of Corollary 1 in the main text without claiming a joint Newton step for the full pipeline.

5 Link Ablation: Offset-Platt vs. Raw Sigmoid

The main text (Section 4.3) reports that offset-Platt outperforms raw $\sigma(\eta_0 + \hat{b}_i)$ on ECE and NLL. Table 6 provides the full breakdown across datasets and link types.

Table 6. Link ablation on AS09 and AS17 (5 backbones × 3 seeds, averaged). **Raw**: $\sigma(\eta_0 + \hat{b}_i)$; **+OP**: offset-Platt $\sigma(a\eta_0 + b_0 + \hat{b}_i)$. Static variants use b_i only; temporal variants add $u_i(t)$.

Dataset	Method	AUC (↑)	NLL (↓)	ECE (↓)
AS09	Platt (no per-item)	0.6782	0.5924	0.0498
	Raw temporal	0.7016	0.6275	0.1071
	Raw temporal + OP	0.7021	0.6052	0.0830
	Raw static	0.7060	0.6206	0.1051
	Static + OP (SLC)	**0.7066**	**0.5956**	**0.0763**
AS17	Platt (no per-item)	0.6814	0.6134	0.0116
	Raw temporal	0.7170	0.5971	0.0317
	Raw temporal + OP	0.7166	0.5951	0.0214
	Raw static	0.7182	0.5955	0.0279
	Static + OP (SLC)	**0.7182**	**0.5931**	**0.0170**

Adding offset-Platt consistently improves ECE and NLL without sacrificing AUC. On AS09: offset-Platt reduces ECE from 10.51% to 7.63% (−2.88 pp) and NLL from 0.621 to 0.596 (−0.025). On AS17: ECE from 2.79% to 1.70% (−1.09 pp) and NLL from 0.596 to 0.593 (−0.002). The offset-Platt link correctly treats $\hat{b}_i$ as a random effect that should not be rescaled by the global parameter a, matching the GLMM parameterization (Corollary 1 in the main text).

6 Calibration-Fraction Sweep

The main text (Section 4.5) reports that ΔAUC scales monotonically with calibration fraction. Table 7 provides the detailed results on AS17 and AS09, averaged over 5 backbones × 3 seeds. All numbers in Table 7 use the final single-pass SLC pipeline.

Table 7. Calibration-fraction sweep on AS17 and AS09 (5 backbones × 3 seeds, averaged). SLC's ΔAUC over Platt scales monotonically with calibration data, confirming signal-driven improvement.

	AS17				AS09			
Calib frac	Platt	SLC	ΔAUC	ΔNLL	Platt	SLC	ΔAUC	ΔNLL
10%	0.6814	0.6938	+1.24	−0.003	0.6782	0.6798	+0.16	+0.008
20%	0.6814	0.7001	+1.87	−0.007	0.6782	0.6830	+0.48	+0.010
50%	0.6814	0.7090	+2.76	−0.013	0.6782	0.6866	+0.84	+0.012
100%	0.6814	0.7182	+3.68	−0.020	0.6782	0.7066	+2.84	+0.003

On AS17, both ΔAUC and ΔNLL improve monotonically as more calibration data become available, confirming that the gain is signal-driven rather than an artifact of a particular split. On AS09, ΔAUC is also monotone, but NLL remains slightly above Platt throughout; the gap shrinks from +0.012 at 50% calibration data to +0.003 at 100%. This matches the main-text conclusion that in extremely sparse regimes, SLC recovers substantial ranking headroom while retaining a small proper-score trade-off against Platt.

7 Flight-Delay and MovieLens Experiments

7.1 Flight-Delay (Positive Control)

The main text (Section 4.6) uses flight-delay as a positive control for the claim that the same phenomenon can arise beyond education when the deployed backbone leaves route-level bias. Table 8 provides the full results across two backbone variants and all methods.

We use US Department of Transportation on-time performance data from 2018–2019 (∼12M flights, ∼2500 routes). We evaluate two backbone variants: bbA (SGDClassifier with carrier/origin/dest features, no per-route parameters) and bbB (the same model family with different regularization). The temporal split uses 2018 for training, early 2019 for calibration, and late 2019 for test, with three random seeds for backbone training.

Table 8. Flight-delay experiment (2 backbones × 3 seeds, mean ± std). Route-aware correction recovers AUC headroom that score-only calibration cannot access; in this dense regime, the less-regularized ResCal baseline attains the highest AUC.

	bbA		bbB	
Method	AUC	NLL	AUC	NLL
Base	0.5813 ± 0.012	0.5511	0.5823 ± 0.012	0.5763
Platt	0.5813 ± 0.012	0.4694	0.5823 ± 0.012	0.4693
Platt-T	0.5734 ± 0.016	0.4811	0.5755 ± 0.014	0.4800
ResCal	0.6121 ± 0.005	0.4847	0.6134 ± 0.006	0.4899
SLC (static+OP)	0.6025 ± 0.007	0.5099	0.6019 ± 0.006	0.5463
SLC +Iso	0.6038 ± 0.011	0.4672	0.6007 ± 0.013	0.4678

Base and Platt produce identical AUC (confirming Lemma 1). Platt-T *decreases* AUC (−0.8 pp on bbA), showing that time-only conditioning without route identity is harmful. Per-route methods (ResCal, SLC) recover +2–3 pp AUC headroom, so the positive-control conclusion is structural rather than method-specific. ResCal achieves the highest AUC, while SLC is slightly more conservative in this dense regime; this matches the main-text density analysis, where shrinkage helps most when per-item calibration data are scarce and can add bias when route-level data are abundant. SLC +Iso achieves the best NLL (0.467–0.468) while preserving the route-aware AUC gain.

7.2 MovieLens-1M (Negative Control)

The main text reports that MovieLens-1M with a matrix-factorization backbone yields $\Delta\,\mathrm{AUC} \approx 0$. Table 9 confirms this.

We use MovieLens-1M (∼1M ratings, 3706 movies, 6040 users) as a binary classification task, with ratings ≥ 4 treated as positive. The two backbones are MF (matrix factorization with per-item embeddings) and NCF (neural collaborative filtering with per-item embeddings). We use a temporal split by timestamp and report means over 3 seeds.

Table 9. MovieLens-1M negative control (3 seeds, mean). When the backbone already models per-item effects, post-hoc per-item correction provides negligible or no benefit.

	MF backbone		NCF backbone	
Method	AUC	NLL	AUC	NLL
Base	0.7949	0.5483	0.7885	0.5682
Platt	0.7949	0.5469	0.7885	0.5548
ResCal	0.7949	0.5481	0.7882	0.5667
SLC	0.7951	0.5469	0.7843	0.5599
Naive	0.7945	0.5507	0.7572	0.6479

On the MF backbone, ΔAUC between Base and SLC is $+0.02$ pp—effectively zero. On the NCF backbone, SLC *decreases* AUC by -0.42 pp, and Naive degrades by -3.13 pp. Together with flight-delay, this indicates that the effect is backbone-relative rather than domain-specific: when the backbone already incorporates item effects via embeddings, post-hoc per-item correction adds noise without recovering headroom. The MovieLens result validates SLC's applicability boundary and demonstrates that the method is appropriately conservative (does not inflate metrics artificially).